

Macroeconomics

Openness in Goods and Financial Markets

(Blanchard, Macroeconomics 8th edition Ch. 17, 19)

Open Economy

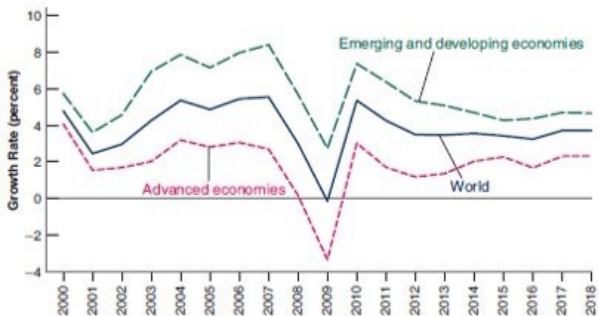
- ▶ Openness has three distinct dimensions: the goods markets, the financial markets, and the factor markets.
- ▶ *Goods* markets: The ability of consumers and firms to choose between domestic goods and foreign goods.
 - ▶ Even countries most committed to free trade have tariffs (taxes on imported goods) and quotas (restrictions on quantity).
 - ▶ Average tariffs have declined dramatically in recent decades.
- ▶ *Financial* markets: The ability of financial investors to choose between domestic assets and foreign assets.
 - ▶ Until recently, even some rich countries had *capital controls*: restrictions on the foreign assets their domestic residents could hold and the domestic assets foreigners could hold.
 - ▶ World financial markets are becoming more closely integrated, but countries such as China still impose capital controls.

Open Economy (con.)

- ▶ Openness in *factor* markets: The ability of firms to choose where to locate production, and of workers to choose where to work
 - ▶ e.g., EU, NAFTA
 - ▶ Immigration in the Europe and US is a hot political issue, especially from low-wage countries.
- ▶ In the short run and in the medium run, openness in factor markets plays much less of a role.
- ▶ Focus on the implications of the first two dimensions of openness.

Worldwide Recession

- ▶ The crisis started in the US, but it affected nearly every country in the world (i.e., worldwide recession)
- ▶ This suggests that economies are *not* closed.



Source: IMF, World Economic Outlook, Oct 2015. Used courtesy of IMF.

Openness in Goods Markets

- ▶ Since 1960, exports and imports have more than tripled in relation to GDP. The US has become a much more open economy.



International Trade

- ▶ The volume of trade is not necessarily a good measure of openness.
- ▶ **Tradable goods:** Goods that compete with foreign goods in either domestic markets or foreign markets.
- ▶ Another index of openness is the proportion of aggregate output composed of tradable goods.
- ▶ For example, firms may retain their domestic market share, but still be exposed to foreign import competition.
- ▶ Tradable goods represent about 60% of aggregate output in the US today.
- ▶ It is at the low end of the range of export ratios compared to other countries.

Export-to-GDP Ratios for OECD Countries, 2017

- ▶ In Denmark, export and import-to-GDP ratios are around 70% and 60%, respectively.

Country	Export Ratio	Country	Export Ratio
United States	12.3%	Germany	47.2%
Japan	16.1%	Austria	53.9%
United Kingdom	28.7%	Switzerland	65.0%
Chile	30.5%	Netherlands	86.4%

- ▶ Does this mean that trade barriers in the US are higher?
- ▶ No, geography and size play a role. e.g., smaller countries tend to specialize in the production of a few products and rely on imports.
- ▶ China's export and import-to-GDP ratios are around 20%.

Can Exports Exceed GDP?

- ▶ Since a country cannot export more than it produces, will the export ratio always be less than one?
- ▶ No, exports may be larger than GDP because exports and imports may include exports and imports of intermediate goods.
- ▶ Example: A country imports intermediate goods (inputs) for \$1 billion. Labor is used to transform them into final goods, paid \$200 million. There are no profits. Suppose \$1 billion worth of final goods is exported. What is the export-to-GDP ratio?
- ▶ In 2017, the ratio of exports to GDP in Singapore was 173%, and in Hong Kong was 189%!

Vocabulary

- ▶ **Nominal exchange rate:** The price of the domestic currency in terms of foreign currency.
- ▶ **Real exchange rate:** The price of domestic goods relative to foreign goods.
- ▶ **(Nominal) appreciation:** An increase in the price of the domestic currency in terms of a foreign currency, i.e., an increase in the exchange rate.
- ▶ **(Nominal) depreciation:** A decrease in the price of the domestic currency in terms of a foreign currency.
- ▶ **Fixed exchange rates:** A system in which two or more countries maintain a constant exchange rate between their currencies.
- ▶ In the fixed exchange rate system, **revaluations (devaluations)** are increases (decreases) in the exchange rate.

Dollar-Pound Nominal Exchange Rate

- ▶ Although the dollar has appreciated relative to the pound over the past four decades, this appreciation has come with large swings in the nominal exchange rate between the two currencies.



- ▶ What happened around Brexit? (Good source of exogenous variation!)

DKK-SEK Exchange Rate

DKK to SEK Chart +34.07% (10Y)

• 1 DKK = 1.58311 SEK Nov 1, 2023, 21:35 UTC

Danish Krone to Swedish Krona



Real Exchange Rate

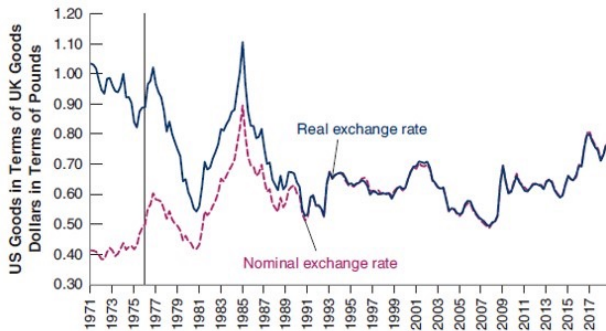
- ▶ The nominal exchange rate is E , e.g., for every dollar, you get E pounds. (Home country is US here.)
- ▶ The real exchange rate, e.g., the price of US goods in terms of British goods, is

$$\varepsilon = \frac{EP}{P^*}$$



- ▶ **Real appreciation (depreciation):** An increase (decrease) in the real exchange rate, i.e., an increase (decrease) in the relative price of domestic goods in terms of foreign goods.

Real Exchange Rate



- Source FRED. GDPDEF, GBRGDPDEFQISM EI, DEXUSUK.

► Except for the difference in trend reflecting higher average inflation in the UK than in the US until the early 1990s, the nominal and the real exchange rates have moved largely together.

Multilateral Real Exchange Rate

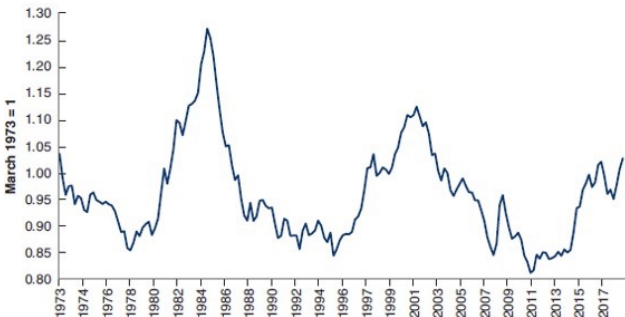
- ▶ How do we go from **bilateral** exchange rates (two countries) to **multilateral** exchange rates? (aka trade-weighted or effective exchange rates)
- ▶ This requires data on the geographic composition of US trade for both exports and imports (for nominal or real exchange rate).

	Percent of Exports to	Percent of Imports from
Canada	18	12
Mexico	16	14
European Union	19	19
China	7	21
Japan	4	6
Rest of Asia and Pacific	15	9
Others	21	19

- ▶ And data on how much US competes in other third countries.

Multilateral Real Exchange Rate (con.)

- ▶ The overall weight for a particular country captures 3 effects
 $\text{import comp} + \text{direct export comp} + \text{third-market comp}$
- ▶ See https://www.bis.org/publ/qtrpdf/r_qt0603e.htm



More Vocabulary

- ▶ **Foreign exchange:** Buying and selling foreign currency.
- ▶ **Balance of payments:** A set of accounts that summarize a country's transactions with the rest of the world.
- ▶ **Current account:** Transactions above the line record payments to and from the rest of the world.
 - ▶ Exports and imports of goods and services (**trade balance**)
 - ▶ Net **income balance** between income received from the rest of the world and income paid to foreigners
 - ▶ **Net transfer received**—the difference in foreign aid given and received
- ▶ **Current account balance:** The sum of net payments to and from the rest of the world. If positive (negative), then surplus (deficit).

More Vocabulary (con.)

- ▶ **Financial account:** Transactions below the line record net foreign holdings of domestic assets
- ▶ **Net capital flows or financial account balance:** An increase in net foreign indebtedness (holdings of domestic assets minus the increase in domestic holdings of foreign assets)
- ▶ **Financial account surplus (deficit):** Positive (negative) net capital flows
- ▶ **Statistical discrepancy:** Difference between current and capital account transactions

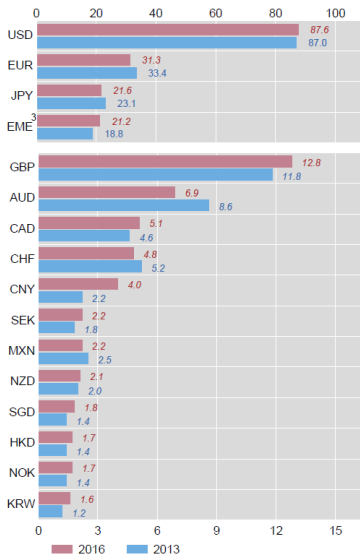
Foreign Exchange Markets

- ▶ The set of markets where foreign currencies and other assets are exchanged for domestic ones
- ▶ Institutions buy and sell deposits of currencies or other assets for investment purposes
- ▶ Daily volume of FX transactions (Apr. 2013): \$ 5.3 trillion USD
- ▶ Up from \$500 billion in 1989 (\$3.3 trillion in 2007)
- ▶ Most transactions (87%/200% in April 2010) exchange foreign currencies for USD
- ▶ Use of USD has increased even more since euro area sovereign debt crisis in 2010 (e.g., Greece)
- ▶ USD is vehicle currency (euro?): greater volume of transactions

Largest Financial Centers for FX (2016)

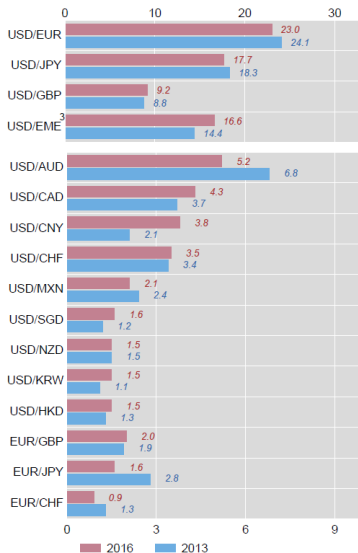
- ▶ Geographical distribution of OTC foreign exchange turnover:
 1. United Kingdom (London): 37.1%
 2. United States (New York): 19.4%
 3. Singapore: 7.9%
 4. Hong Kong: 6.7%
 5. Japan (Tokyo): 6.1%
 6. Switzerland: 2.4%

FX Market Turnover



Source: Triennial Central Bank Survey 2016, BIS

FX Market Turnover, By Pair



Source: Triennial Central Bank Survey 2016. BIS

FX markets

- ▶ Buying/selling in FX markets are dominated by commercial and investment banks
- ▶ Central banks sometimes intervene, but direct effects of their transactions are small and transitory in many countries
- ▶ Computer and telecommunications technology transmit information rapidly and have integrated markets
- ▶ Integration of financial markets implies that there can be no significant differences in exchange rates across locations
- ▶ Arbitrage: buy at low price and sell at higher price for a profit
- ▶ If euro sells for \$1.1 in New York and \$1.2 in London, could buy euros in New York (cheaper) and sell in London at a profit

US Balance of Payments (BOP), 2018

Particulars	Billions of U.S. Dollars	Billions of U.S. Dollars
Current Account		
Exports	2,500	
Imports	3,122	
Trade balance (deficit = minus sign) (1)		-622
Income received	1,200	
Income paid	1,067	
Net income (2)		133
Current account balance (1) + (2) (deficit = minus sign)		-489
Financial Account		
Net capital transfers (3)	9	
Increase in foreign holdings of US assets (4)	811	
Increase in US holdings of foreign assets (5)	301	
Financial account balance (7) = (3) + (4) - (5)		519
Statistical discrepancy		30
financial account - current account balance		

Source: US Bureau of Economic Analysis, US International Transactions, Table. 17.1.

Openness in Financial Markets

- ▶ Gross domestic product (GDP) measures value added domestically.
- ▶ **Gross national product (GNP)** measures the value added by domestic factors of production

$$GNP = GDP + NI$$

- ▶ where NI denotes net income—payments received from the rest of the world less income paid to the rest of the world.
- ▶ In most countries, the difference between GNP and GDP is small (relative to GDP).

GDP versus GNP: The Example of Kuwait

- ▶ Kuwait discovered oil, and the government decided to save a portion of oil revenues for future generations.
- ▶ Kuwait ran a large current account surplus and accumulated large foreign assets, resulting in larger GNP compared to GDP.

Year	GDP	GNP	Net Income (NI)
1989	7,143	9,616	2,473
1990	5,328	7,560	2,232
1991	3,131	4,669	1,538
1992	5,826	7,364	1,538
1993	7,231	8,386	1,151
1994	7,380	8,321	941

- Source: *International Financial Statistics, IMF*. All numbers are in millions of Kuwaiti dinars. 1 dinar = \$3.0. (2018).

Domestic vs Foreign Assets

Uncovered Interest Parity

- ▶ Arbitrage implies that:

$$1 + i_t = (1 + i_t^*) \left(\frac{E_t}{E_{t+1}^e} \right)$$

- ▶ which is called the **uncovered interest parity** relation or the interest parity relation.
- ▶ The assumption that financial investors hold only the bonds with the highest expected rate of return ignores transactions costs and risk.

Uncovered Interest Parity

- ▶ We can rewrite as

$$1 + i_t = \frac{(1 + i_t^*)}{1 + \frac{E_{t+1}^e - E_t}{E_t}}$$

- ▶ which gives a good approximation of the interest parity condition:

$$i_t \approx i_t^* - \frac{E_{t+1}^e - E_t}{E_t}$$

- ▶ The condition states that the domestic interest rate must be equal to the foreign interest rate minus the expected depreciation of the foreign currency.
- ▶ Why? Arbitrage by investors implies that the domestic interest rate must be equal to the foreign interest rate minus the expected appreciation rate of the domestic currency.

Three-Month Nominal Interest Rates: US and UK



- Source: FRED TB3MS; IR3TTS01GBM156N

Buying Brazilian Bonds

- ▶ In September 1993 Brazilian bonds were paying a monthly interest rate of 36.9% while US bonds paid 0.2% monthly.
- ▶ However, the Brazilian currency (cruzeiro) was depreciating rapidly. It fell 34.6% in August 1993 versus the dollar.
- ▶ When you factor in the currency movements the expected rate of return in dollars from holding Brazilian bonds is only $1.017 - 1 = 1.7\%$ per month.
- ▶ While this return is higher you need to consider risk and transaction costs.

Spot Rates and Forward Rates

- ▶ **Spot rates:** exchange rates for currency exchanges “on the spot,” or when trading is executed in the present (within two days)
- ▶ **Forward rates:** exchange rates for currency exchanges that will occur at a future (“forward”) date
- ▶ Forward dates typically 30, 90, 180, or 360 days in the future
- ▶ Rates negotiated between two parties in present, but exchange occurs in future
- ▶ Forward rate is set by contract today
- ▶ Financial derivatives: forwards, swaps, futures, options

Dollar/pound Spot and Forward Exchange Rates



Source: Datastream. Rates shown are 90-day forward exchange rates and spot exchange rates, at end of month.

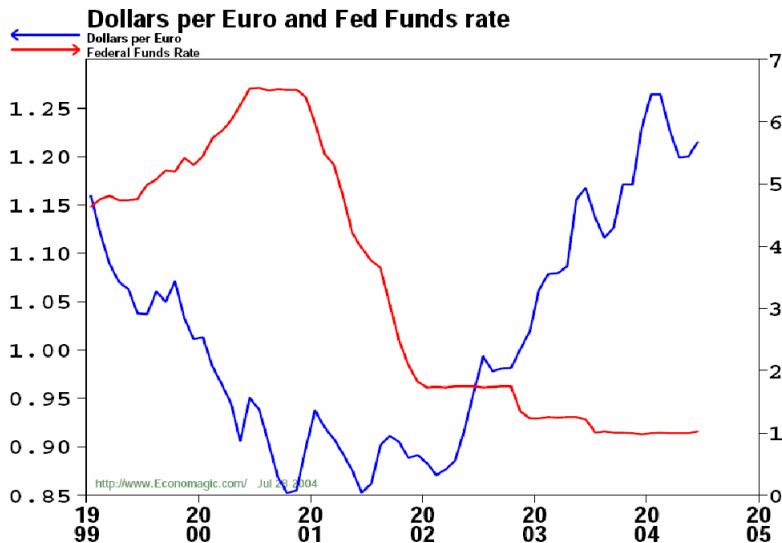
Example of Forward Rate

- ▶ You have to pay Japanese supplier (e.g., of cars) in 30 days
- ▶ Can sell car at 340,000 DKK; pay supplier 5M ¥
- ▶ Spot exchange rate: $0.05 \text{ DKK/¥} \Rightarrow \text{pay } 5\text{M ¥} \times 0.05 \text{ DKK/¥} = 250,000 \text{ DKK per car} \Rightarrow \text{earn } 340,000 - 250,000 = 90,000 \text{ DKK per car}$
- ▶ Suppose you don't have the money right now, need to sell the cars first
- ▶ In 30 days, DKK unexpectedly depreciates to 0.07 DKK/¥
- ▶ You will then pay 350,000 DKK $\Rightarrow$ *lose* 10,000 per car

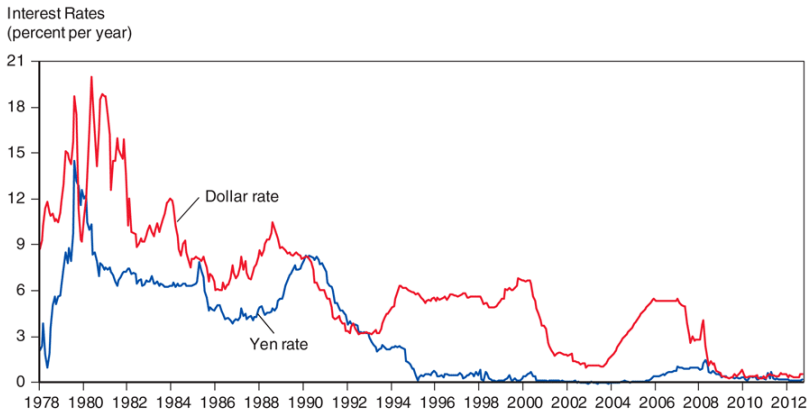
Use of forward rate

- ▶ To avoid this risk, you can make a 30-day FX deal with a bank
- ▶ Forward rate 0.065 DKK/¥
- ▶ Guaranteed to pay $0.065 \text{ DKK/¥} \times 5\text{M ¥} = 325,000 \Rightarrow$ profit of 15,000 DKK per car
- ▶ You have **hedged** your foreign currency risk
- ▶ Note: if DKK depreciates more in these 30 days, you “win”; if DKK appreciates, you “lose”

Evidence of UIP?



Interest rates: dollar vs. yen deposits



Source: Datastream. Three-month interest rates are shown.

Carry trade

- ▶ Perhaps risk and liquidity matter more than we think
- ▶ UIP tells us to flip a coin since returns are equalized
- ▶ Carry trade: borrow low-interest currencies and buy high-interest currencies to profit
- ▶ If UIP holds, carry trade would not exist
- ▶ Strategy of going short in a low-interest rate currency (e.g., Japanese yen, Swiss franc, USD, Euro), and simultaneously long in a high-interest rate currency (e.g., New Zealand dollar, Australia, Brazil, Turkey)

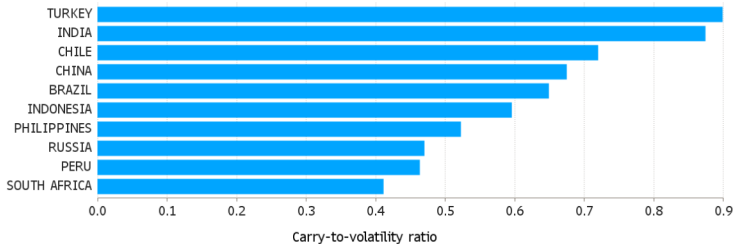
Carry trade (Frankel)

- ▶ Carry trade pays off “on average”
- ▶ Carry trade hazard: “investment” currency (low-interest) may experience abrupt crashes
- ▶ But volatility is high, can produce big losses: “unwinding of the carry trade”
- ▶ <https://www.bloomberg.com/news/articles/2016-05-02/turkey-s-lira-leads-carry-trade-revival-as-investors-ignor>
- ▶ <https://www.youtube.com/watch?v=1BQiV9BPFNs>

Carry trade: Turkish lira

Turkey Carries On

Lira's volatility-adjusted yields rank highest among major currencies

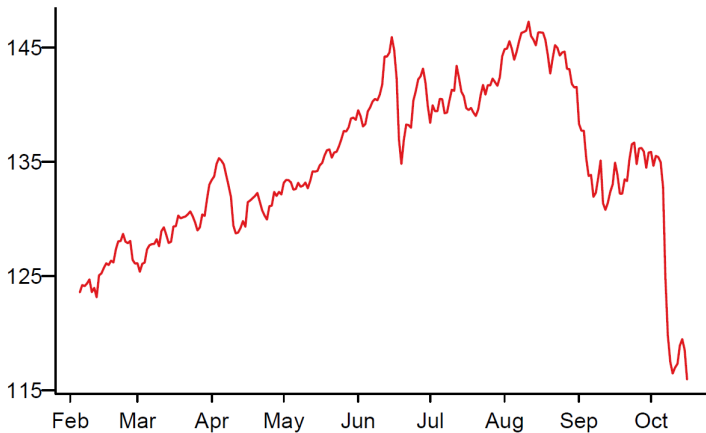


Source: Bloomberg

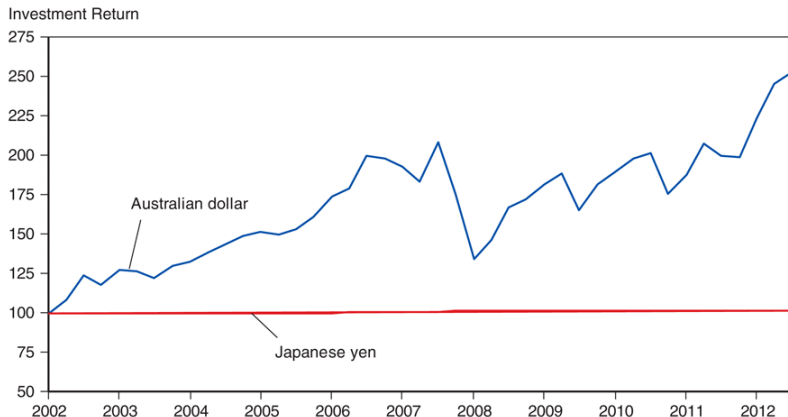
Calculation is based on three-month implied yields in currency forwards and implied volatility as of April 29.

Bloomberg 

Yen vs USD (1998)



Yen vs Australian dollar



Source: Exchange rates and three-month treasury yields from Global Financial Data.

Cumulative total investment return

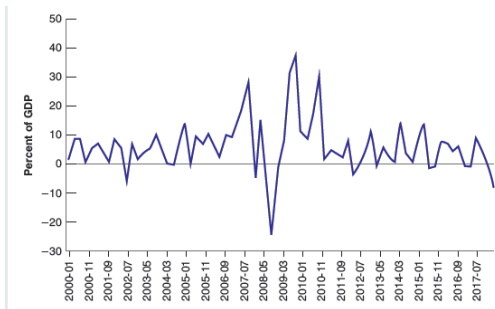
Why UIP might not hold

- ▶ Risk premium: compensate investors for uncertainty about x-rate (i.e., return is higher than suggested by UIP)
- ▶ Expectational errors
- ▶ Investors are slow to react

Capital Flows and Sudden Stops

- ▶ Increased risk in assets may lead investors to sell, regardless of the interest rate.
- ▶ Again, there may be limits to the interest parity condition.
- ▶ Many Latin American and Asian emerging economies have experienced *sudden stops*: selling episodes in which the exchange rate falls a lot without much change in domestic or foreign interest rates.
- ▶ Investors may switch towards *safe havens*.
- ▶ Large capital outflows from several emerging economies during Great Recession.

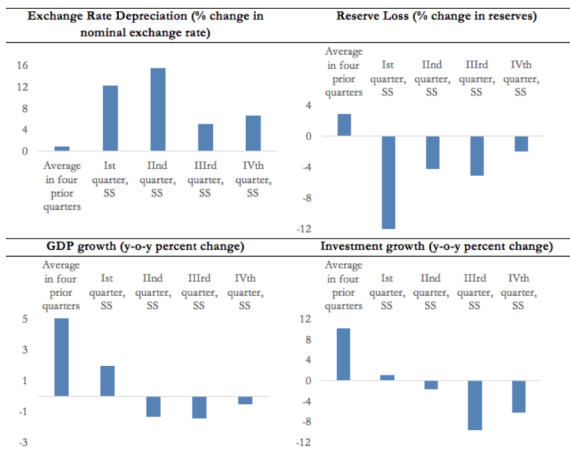
Capital Flows and Sudden Stops (con.)



Source: IMF BOP statistics.

- ▶ Downward pressure on x-rates, problems in financial systems
- ▶ Domestic banks relying on foreign investors cut lending.
- ▶ Transmission of crisis from US to world.

Financial and Real Impact of Sudden Stops



“Managing sudden stops”, Eichengreen and Gupta.

Equilibrium in the Goods Market

- ▶ We now introduce the **Mundell-Fleming model**, an extension to the open economy of the IS-LM model.
- ▶ In an open economy, the demand for domestic goods is

$$Z = C + I + G - \frac{IM}{\varepsilon} + X$$

- ▶ Imports are positively related to domestic income (Y) and the real exchange rate (ε)

$$\oplus \oplus$$

$$IM = IM(Y, \varepsilon)$$

- ▶ Exports are positively related to foreign income (Y^*) and negatively related to ε

$$\oplus \ominus$$

$$X = X(Y^*, \varepsilon)$$

Depreciation, the Trade Balance, and Output

- ▶ Recall the real exchange rate equation:

$$\varepsilon = \frac{EP}{P^*}$$

- ▶ and the definition of net exports:

$$NX = X - \frac{IM}{\varepsilon}$$

- ▶ Then, we have

$$NX = X(Y^*, \varepsilon) - \frac{IM(Y, \varepsilon)}{\varepsilon}$$

Depreciation, the Trade Balance, and Output

- ▶ Real depreciation ($\varepsilon \downarrow$) affects the trade balance through three separate channels:
 - ▶ Exports, X , increase.
 - ▶ Imports, IM , decrease.
 - ▶ The relative price of foreign goods in terms of domestic goods, $1/\varepsilon$, increases.
- ▶ **Marshall-Lerner condition:** A real depreciation leads to an increase in net exports.
- ▶ Depreciation leads to a shift in demand, both foreign and domestic, toward domestic goods.
- ▶ The shift in demand leads, in turn, to both an increase in domestic output and an improvement in their trade balance.

Deriving the Marshall-Lerner Condition

- ▶ We do this under the assumption of balanced trade.

$$\begin{aligned}NX &= X - \frac{IM}{\varepsilon} \\ \Rightarrow \varepsilon NX &= \varepsilon X - IM \\ \Rightarrow (\Delta \varepsilon)NX + \varepsilon(\Delta NX) &= (\Delta \varepsilon)X + \varepsilon(\Delta X) - (\Delta IM) \\ \Rightarrow \frac{\varepsilon(\Delta NX)}{\varepsilon X} &= \frac{(\Delta \varepsilon)X}{\varepsilon X} + \frac{\varepsilon(\Delta X)}{\varepsilon X} - \frac{(\Delta IM)}{\varepsilon X} \\ \Rightarrow \frac{(\Delta NX)}{X} &= \frac{(\Delta \varepsilon)}{\varepsilon} + \frac{(\Delta X)}{X} - \frac{(\Delta IM)}{IM} > 0\end{aligned}$$

- ▶ The Marshall-Lerner condition is the condition that the sum of these three terms be positive.
- ▶ If it is satisfied, a real depreciation leads to an improvement in the trade balance.

Mundell-Fleming Model

- ▶ Because $Y = Z$, the equilibrium condition for output can be expressed as:

$$Y = C \overset{(+)}{(Y - T)} + I \overset{(+)}{(Y, r)} \overset{(-)}{(r)} + G + NX \overset{(-)}{(Y, Y^*, \varepsilon)} \overset{(+)}{(Y^*)} \overset{(-)}{(\varepsilon)}$$

- ▶ Focus on the short run, therefore, P and P^* are fixed. Further simplify by setting $P = P^*$, so $\varepsilon = E$.

$$Y = C \overset{(+)}{(Y - T)} + I \overset{(+)}{(Y, r)} \overset{(-)}{(r)} + G + NX \overset{(-)}{(Y, Y^*, E)} \overset{(+)}{(Y^*)} \overset{(-)}{(E)}$$

- ▶ Goods market equilibrium implies that output depends negatively on both the nominal interest rate and the nominal exchange rate.

Mundell-Fleming Model (con.)

- ▶ Recall UIP, and assume that expected future exchange rate is given:

$$1 + i = (1 + i^*) \frac{E}{\bar{E}^e} \Rightarrow E = \frac{1 + i}{1 + i^*} \bar{E}^e$$

- ▶ The open economy versions of the IS and LM relations:

$$IS : Y = C(Y - T) + I(Y, i) + G + NX \left(Y, Y^*, \frac{1 + i}{1 + i^*} \bar{E}^e \right)$$

$$LM : i = \bar{i}$$

- ▶ An increase in the interest rate now has two effects:
 - ▶ The direct effect on investment
 - ▶ The secondary effect through the exchange rate

Effects of Monetary Policy in an Open Economy

Effects of Fiscal Policy in an Open Economy

Policy Trilemma

1. Exchange rate stability

- ▶ May be desired as a means to promote stability in trade and investment; represented by zero expected depreciation

$$\frac{E^e - E}{E} = 0$$

2. International capital mobility

- ▶ May be desired a means to promote integration, efficiency, and risk sharing; represented by UIP (results from arbitrage)
- ▶ <https://www.youtube.com/watch?v=hxSm1A-zBxA>

$$i = i^* - \frac{E^e - E}{E}$$

3. Monetary policy autonomy

- ▶ May be desired as means to manage Home economy's business cycle; represented ability to set Home interest rate independently of Foreign interest rate

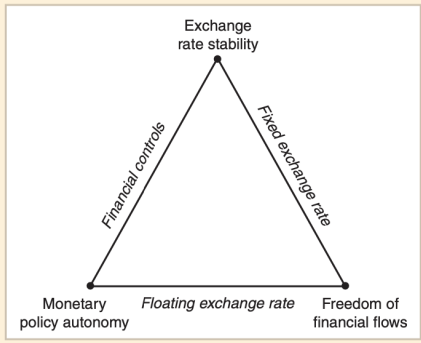
$$i \neq i^*$$

Policy Trilemma (cont.)

Figure 19-1

The Policy Trilemma for Open Economies

The vertices of the triangle show three features that policy makers in open economies would prefer their monetary system to achieve. Unfortunately, at most two can coexist. Each of the three policy regime labels along the triangle's edges (floating exchange rate, fixed exchange rate, financial controls) is consistent with the two goals that it lies between in the diagram.

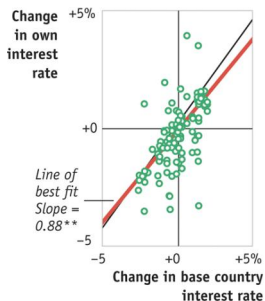


Monetary Autonomy and Stability

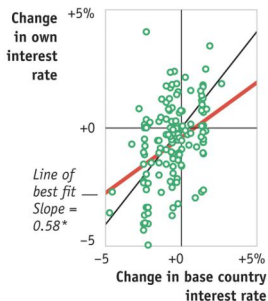
- ▶ Do fixed exchange rates diminish monetary autonomy and stability?
- ▶ When a country pegs, it relinquishes independent monetary policy
- ▶ Must adjust money supply M^s at all times to ensure that home interest rate i equals foreign rate i^* (plus any risk premium)
- ▶ To solve the trilemma, a country can do the following:
 1. Opt for open capital markets, with fixed x-rates (“open peg”)
 2. Opt to open its capital market but allow the currency to float (“open nonpeg”)
 3. Opt to close its capital markets (“closed”)
- ▶ <https://www.youtube.com/watch?v=FdwPD2JbdJo>

Trilemma in Action

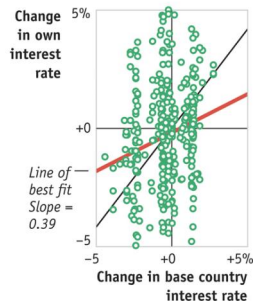
(a) Open and Pegged



(b) Open and Not Pegged



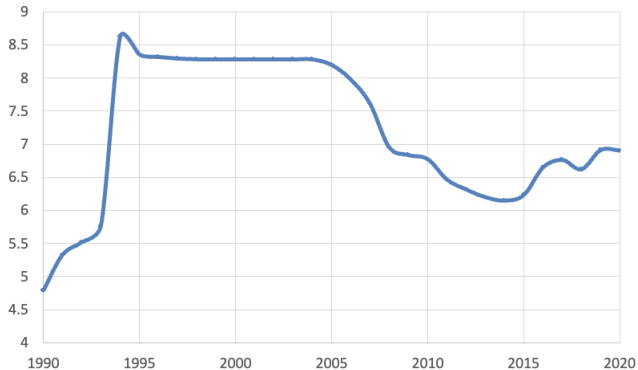
(c) Closed



RMB Internationalization

- ▶ China joins the WTO in 2001, and the current account surplus further widened, from 2% of GDP in 2000 to 10.6% in 2007.
- ▶ This led to a growing injection of money into the domestic economy and pressures of currency appreciation.
- ▶ To maintain a fixed exchange rate, PBOC tried to sterilize the external source of money supply by *increasing* the reserve requirement (reducing the money multiplier) and *issuing* bills in open market operations.
- ▶ To retain independent control of monetary policy decisions for long-term interests, China adopted a more flexible exchange rate regime: “a managed floating exchange rate based on market supply and demand with reference to a basket of currencies”.
- ▶ <https://www.bis.org/review/r100803c.pdf>

RMB Internationalization



RMB to USD exchange rate.

Fixed Exchange Rates

- ▶ At one end of the spectrum are countries with flexible exchange rates such as the US, the UK, Japan, and Canada.
 - ▶ No explicit exchange rate targets.
- ▶ At the other end are countries that operate under fixed x-rates.
 - ▶ Maintain a fixed exchange rate in terms of some foreign currency; for example, some peg their currency to the dollar.
 - ▶ eg., from 1991 to 2001, Argentina pegged its currency, the peso, at the highly symbolic exchange rate of one dollar for one peso
- ▶ Some countries operate under a crawling peg by moving to an exchange rate target slowly.
 - ▶ Typically have inflation above US inflation, so a peg would lead to real appreciation.
 - ▶ Instead, choose a predetermined rate of depreciation against the dollar, e.g., Vietnam.

Fixed Exchange Rates: Monetary Policy

- ▶ The difference between fixed and flexible exchange is the ability of the central bank to respond.

$$1 + i_t = (1 + i_t^*) \frac{E_t}{E_{t+1}^e}$$

- ▶ If the country pegs the exchange rate so that $E_t = \bar{E}$, which also is the expectation of the future exchange rate, then the interest rate relation becomes:

$$1 + i_t = (1 + i_t^*) \Rightarrow i_t = i_t^*$$

- ▶ Under a fixed exchange rate and perfect capital mobility, the domestic interest rate must be equal to the foreign interest rate.
- ▶ Under fixed exchange rates, the central bank gives up monetary policy as a policy instrument.